1) Fraud Transaction detection (linear search)

	1201	1305	1450	1408	1502
i =	0	1	2	3	4

find fraudId.

Solⁿ :-

```
int fraudId; cin >> fraudId;
for (int i=0; i<N; i++) {
    if (arr[i] == fraudId) {
     cout << "Fraud Transaction found" << endl;
    } else {
     cout << "Transaction not found" << endl;
    }
    return;
}
```

2) Server log search (Binary Search).

	1001	1005	1010	1015	1020	1030	1040
i =	low 0	1	2 ~~mid~~	3 mid	4	5	6 high

target = 1015

```
while (low <= high) {
  mid = low + (high - low)/2

  if (arr[mid] == target) { cout << "log found" << endl;
                              return;
                          }
  else if (arr[mid] < target) { low = mid+1 ; }

  else { high = mid -1 ;
  }
}
cout << "log not found" << endl;
}
```

Salaries (Bubble Sort).

___/___/___

③ Sort Employee Salaries (Bubble Sort).

i = 0 to N-1

j = 1 to N-i-1

Pass-1

j= 1	45000	32000	55000	28000	60000
	arr[j]	swap	arr[j+1]		
j	arr[j] > arr[j+1]				
j=2	32000	45000	55000	28000	60000
		no swap			
j=3	32000	55000	45000	28000	60000
			swap		
j=4	32000	55000	28000	45000	60000 → fixed (largest element)
j=5	32000	55000	28000	45000	swap 60000 sorted

Pass-2

j	32000	55000	28000	45000	60000
	do not swap				
j=2	32000	55000	28000	45000	60000
		swap			
j=3	32000	28000	55000	45000	60000
			swap		
j=4	32000	28000	45000	55000	60000

Pass-3

j=1	32000	28000	45000	55000	60000
	swap				
j=2	28000	32000	45000	55000	60000
		do not swap			
j=3	28000	32000	45000	55000	60000

Pass-4

j=1	28000	32000	45000	55000	60000
	do not swap				
j=2	28000	32000	45000	55000	60000

```
for (i=0 ; i<N ; i++) {
  for (j=0 ; j<N-i-1 ; j++) {
    if (arr[j] > arr[j+1]) {
      int temp = arr[j];
      arr[j] = arr[j+1];
      arr[j+1] = temp;
    }
  }
}
```

_/_/_

④ Sort website response time (Insertion Sort).

250 120 320 180 100

Compare j & j++ :- check for elements & insert it at correct place.

250 | 120 320 180 100

120 250 | 320 180 100

120 250 320 | 180 100

120 180 250 320 | 100

100 120 180 250 320 (sorted)

```
for (int i=1; i<N; i++) {
    int key = arr[i];
    int j = i-1;
    while (j>=0 && arr[j] > key) {
        arr[j+1] = arr[j];
        j--;
    }
    arr[j+1] = key;
}
```

~~cout << arr~~ print sorted array.

⑤ Bug Reports (Selection Sort) :-

9 2 8 1 5

1 9 2 8 5

1 2 9 8 5

1 2 5 9 8

1 2 5 8 9

```
for (i=0; i<N-1; i++) {
    mini = i;
    for (j = i+1; j<N; j++) {
        if (arr[j] < arr[mini]) {
            mini = j;
        }
    }
    temp = arr[i];
    arr[i] = arr[mini];
    arr[mini] = temp;
}
```

print array.

fraudTransac

```
#include <iostream>
using namespace std;

bool findFraudId(int arr[], int N, int fraudId){
 for(int i=0; i<N; i++){
    if(arr[i]==fraudId)
        return true;
 }
     return false;
}

int main(){
    int N, fraudId;
    cin>>N;
    int arr[100];

    for(int i=0; i<N; i++){
        cin>>arr[i];
    }
    cin>>fraudId;

    findFraudId(arr,N,fraudId);

    if(findFraudId(arr,N,fraudId)){
    cout<<"Fraud Transaction found"<<endl;
    }else{
    cout<<"Transaction not found"<<endl;
    }
    return 0;

}
```

Search

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```
if(findFraudId(arr,N,fraudId)){
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powe

```
PS C:\PD\DSA\Searching> g++ fraudTransactionDetection.cpp
PS C:\PD\DSA\Searching> ./a.exe
5
1201
1305
1450
1408
1502
1450
Fraud Transaction found
PS C:\PD\DSA\Searching>
```

rverLogSearch.cpp

```cpp
#include <iostream>
using namespace std;

void findServerLog (int arr[], int N, int target){
  int low=0, high=N-1;
  while(low <= high){
    int mid=low+(high-low)/2;
    if(arr[mid]==target){
        cout<<"Log Found"<<endl;
        return;
    }else if(arr[mid]<target){
        low=mid+1;
    }else{
        high=mid-1;
    }
  }
  cout<<"Log Not Found"<<endl;
}

int main() {
    int N, target;
    cin>>N;
    int arr[1000];
    for(int i=0; i<N; i++){
        cin>>arr[i];
    }
    cin>>target;
    findServerLog(arr,N,target);
    return 0;
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   powershell

```
PS C:\PD\DSA\Searching> ./a.exe
7
1001
1005
1010
1015
1020
1030
1040
1015
Log Found
PS C:\PD\DSA\Searching>
```

ogSearch.cpp employeeSalaries.cpp × bugReports.cp

```cpp
#include <iostream>
using namespace std;

void bubbleSort(int arr[], int N){
    for(int i=0; i<N; i++){
        for(int j=0; j<N-i-1; j++){
            if(arr[j] > arr[j+1]) {
                int temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    for(int i=0; i<N; i++){
        cout<<arr[i]<<" ";
    }
    return;
}

int main() {
    int N;
    cin>>N;
    int arr[1000];
    for(int i=0; i<N; i++) {
        cin>>arr[i];
    }
    bubbleSort(arr,N);
    return 0;
}
```

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PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
PS C:\PD\DSA\Searching> g++ employeeSalaries.cpp
PS C:\PD\DSA\Searching> ./a.exe
5
45000
32000
55000
28000
60000
28000 32000 45000 55000 60000
PS C:\PD\DSA\Searching>
```

```cpp
#include <iostream>
using namespace std;
void insertionSort(int arr[], int N){
    for(int i=1; i<N; i++){
        int key=arr[i];
        int j=i-1;
        while (j>=0 && arr[j]>key){
            arr[j+1]=arr[j];
            j--;
        }
        arr[j+1]=key;
    }
    for(int i=0; i<N; i++){
        cout<<arr[i]<<" ";
    }
    return;
}
int main () {
    int N;
    cin>>N;
    int arr[1000];
    for(int i=0; i<N; i++) {
        cin>>arr[i];
    }
    insertionSort(arr,N);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\PD\DSA\Searching> g++ websiteResponse.cpp
PS C:\PD\DSA\Searching> ./a.exe
5
250
120
320
180
100
100 120 180 250 320
PS C:\PD\DSA\Searching>
```

bugReports.cpp ×

```cpp
#include <iostream>
using namespace std;
void selectionSort(int arr[], int N){
    int i,j, mini, temp;
    for(i=0; i<N-1; i++){
        mini=i;
    for(j=i+1; j<N; j++){
        if (arr[j]< arr[mini]){
            mini=j;
        }
    }
    temp=arr[i];
    arr[i]=arr[mini];
    arr[mini]=temp;
    }
     for(int i=0; i<N; i++){
        cout<<arr[i]<<" ";
    }
    return;
}
int main () {
    int N;
    cin>>N;
    int arr[1000];
    for(int i=0; i<N; i++) {
        cin>>arr[i];
    }
    selectionSort(arr,N);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\PD\DSA\Searching> g++ bugReports.cpp
PS C:\PD\DSA\Searching> ./a.exe
5
9
2
8
1
5
1 2 5 8 9
PS C:\PD\DSA\Searching>
```

Search